National Institute for Space Research (INPE)

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Mathematical Methods of Physics I

Exercise Set for Exam 1

Lectures 1–13

Prof. Dr. José Paulo Marchezi

Space Weather Division (DICEP)
National Institute for Space Research (INPE)

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How to Use This Exercise Set

This exercise set covers the material developed in **Lectures 1–13**, corresponding to the content assessed in Exam 1.
The problems are intentionally varied: some reinforce definitions and computational techniques, while others require proofs, connections between different topics, or applications in mathematical and space physics.

Exam 1: November 17, 2026.
The submission deadline, format, and grading weight will be announced separately.

Part	Main topics
I	Matrices, vector spaces, linear operators, and change of basis
II	Inner products, orthogonality, projections, and Hilbert spaces
III	Eigenvalues, diagonalization, spectral theory, and matrix functions
IV	Vector calculus and tensor algebra
V	Functions of a complex variable

1. Matrices, Vector Spaces, and Linear Operators

1.1. Linear Combinations, Bases, and Linear Systems

1.1 Linear combination and span. Let

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \qquad \mathbf{a}_2 = \begin{pmatrix} 5 \\ -4 \\ -7 \end{pmatrix}, \qquad \mathbf{a}_3 = \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} -4 \\ 3 \\ h \end{pmatrix}.$$

Determine the condition on $h \in \mathbb{R}$ for which

$$\mathbf{b} \in \operatorname{span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}.$$

State whether the representation of $\mathbf{b}$ as a linear combination is unique whenever it exists.

1.2 Linear independence depending on a parameter. For

$$\mathbf{v}_1 = (\chi, -1, -1), \qquad \mathbf{v}_2 = (-1, \chi, -1), \qquad \mathbf{v}_3 = (-1, -1, \chi),$$

determine all $\chi \in \mathbb{R}$ for which $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent. Interpret your result in terms of the determinant of the matrix having these vectors as columns.

1.3 Balancing equations as a null-space problem. Use a vector-equation or matrix approach to balance

$$\mathrm{Al_2O_3 + C \longrightarrow Al + CO_2}.$$

Write the stoichiometric coefficients as an unknown vector, construct the homogeneous linear system expressing conservation of each chemical element, and identify the role of the null space.

1.4 Steady-state heat equation on a discrete grid. Four unknown interior temperatures T_1, T_2, T_3, T_4 form a 2×2 grid. Each interior temperature equals the arithmetic mean of its four nearest neighbors. The boundary temperatures adjacent to the grid are:

	upper left	upper right
top boundary	20°C	20°C
bottom boundary	30°C	30°C

and the left and right boundaries are respectively 10°C and 40°C at both levels.

(a) Write the four averaging equations.

(b) Express them as $A\mathbf{T} = \mathbf{b}$.

(c) Solve for the four interior temperatures.

(d) Explain why this is a discrete version of a boundary-value problem.

1.5 Trace identities. For $A, B \in \mathcal{M}_{n\times n}(\mathbb{K})$ and $\alpha \in \mathbb{K}$, prove directly from the definition of trace that

$$\operatorname{Tr}(A+B) = \operatorname{Tr}(A) + \operatorname{Tr}(B), \qquad \operatorname{Tr}(\alpha A) = \alpha \operatorname{Tr}(A),$$
$$\operatorname{Tr}\left(A^T\right) = \operatorname{Tr}(A), \qquad \operatorname{Tr}(AB) = \operatorname{Tr}(BA).$$

Then prove that

$$\operatorname{Tr}\left(S^{-1}AS\right) = \operatorname{Tr}(A)$$

for every invertible S, and explain why trace is therefore independent of the basis used to represent a linear operator.

1.6 Determinant, rank, kernel, and image. Consider $T : \mathbb{R}^3 \to \mathbb{R}^3$, $T(\mathbf{x}) = A\mathbf{x}$, with

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

(a) Compute $\det(A)$. Is A invertible?

(b) Use Gaussian elimination to determine $\operatorname{rank}(A)$.

(c) Use the rank–nullity theorem to determine $\dim(\ker A)$ and find a basis for $\ker(A)$.

(d) For $\mathbf{b} = (b_1, b_2, b_3)^T$, use the Rouché–Capelli theorem to find the necessary and sufficient condition for $A\mathbf{x} = \mathbf{b}$ to be consistent.

(e) Determine $\dim(\operatorname{Im} T)$ and give a basis for $\operatorname{Im} T$.

(f) Describe geometrically the set $\operatorname{Im} T \subset \mathbb{R}^3$.

1.7 Homogeneous system and nontrivial solutions. Consider

$$\begin{cases} \alpha x + y + z = 0, \\ x + \beta y + z = 0, \\ x + y + \alpha\beta z = 0. \end{cases}$$

Determine the relation among α and β for which the system has a nontrivial solution. Your argument must explicitly use the determinant of the coefficient matrix.

1.8 Basis and coordinates in $\mathbb{R}^4$. Let

$$\begin{aligned} \mathbf{v}_1 &= (1, 2, 4, -2), & \mathbf{v}_2 &= (1, 0, 3, -2), \\ \mathbf{v}_3 &= (1, 1, -3, 5), & \mathbf{v}_4 &= (-1, 0, -3, 1), \\ \mathbf{v}_5 &= (-1, 4, -2, 7). \end{aligned}$$

(a) Show that $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is a basis of $\mathbb{R}^4$.

(b) Find $[\mathbf{v}_5]_{\mathcal{B}}$.

(c) Explain why the coordinates are unique.

1.9 Change of basis in a function space. Let

$$V = \operatorname{span}\{\sin x, \cos x\}, \qquad \mathcal{B} = \{\sin x, \cos x\},$$

and define

$$g_1 = 2\sin x + \cos x, \qquad g_2 = 3\cos x.$$

(a) Show that $\mathcal{B}' = \{g_1, g_2\}$ is a basis of V.

(b) Determine the change-of-coordinate matrices $P_{\mathcal{B}\leftarrow\mathcal{B}'}$ and $P_{\mathcal{B}'\leftarrow\mathcal{B}}$.

(c) For

$$h(x) = 2\sin x - 5\cos x,$$

find $[h]_{\mathcal{B}}$ and $[h]_{\mathcal{B}'}$.

(d) Verify explicitly that both coordinate vectors represent the same function.

1.10 A linear operator on polynomials. Let P_2 be the space of polynomials of degree at most two, with $\mathcal{B} = \{1, x, x^2\}$. Define

$$(Tp)(x) = p(2x - 5).$$

(a) Verify that $T : P_2 \to P_2$ is linear.

(b) Find $[T]_{\mathcal{B}}$.

(c) For

$$p_0(x) = 1 + 2x + 5x^2,$$

compute $[T(p_0)]_{\mathcal{B}}$ by matrix multiplication.

(d) Verify the result by direct substitution into $p_0(2x - 5)$.

1.11 Active and passive transformations. Let

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

For $\mathbf{v} = (2,1)^T$:

(a) compute the components after an *active* rotation of the vector by $\theta = \pi/6$;

(b) compute the components of the unchanged vector in a basis rotated counterclockwise by $\pi/6$;

(c) explain why the two operations involve inverse matrices.

1.12 Dual basis and covectors. In $\mathbb{R}^2$, let

$$\mathbf{e}'_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{e}'_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

(a) Find the dual basis $\{\mathbf{e}'^1, \mathbf{e}'^2\}$.

(b) Verify $\mathbf{e}'^i(\mathbf{e}'_j) = \delta^i{}_j$.

(c) Let $\omega = 3\,dx - 2\,dy$. Determine its components in the dual basis.

(d) Verify that the scalar $\omega(\mathbf{v})$ is basis-independent for $\mathbf{v} = (1,-1)^T$.

2. Inner Products, Orthogonality, Projections, and Hilbert Spaces

2.1 Orthogonality, norm, and angle. Let

$$\mathbf{x}_1 = (1,3,-1), \qquad \mathbf{x}_2 = (0,2,4), \qquad \mathbf{x}_3 = (2,1,-1).$$

Find $\alpha \in \mathbb{R}$ such that

$$\mathbf{x}_1 + \alpha\mathbf{x}_2 \perp \mathbf{x}_3.$$

Then determine

$$\|\mathbf{x}_1 + \alpha\mathbf{x}_2\|$$

and the angle between $\mathbf{x}_1 + \alpha\mathbf{x}_2$ and $\mathbf{x}_2$.

2.2 Projection and physical decomposition. Consider

$$\mathbf{B} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \qquad \mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

(a) Normalize $\mathbf{n}$ to obtain $\hat{\mathbf{n}}$ and compute $\|\mathbf{B}\|$.

(b) Determine

$$\mathbf{B}_\parallel = \langle \hat{\mathbf{n}}, \mathbf{B} \rangle \hat{\mathbf{n}}, \qquad \mathbf{B}_\perp = \mathbf{B} - \mathbf{B}_\parallel.$$

(c) Verify $\langle \mathbf{B}_\parallel, \mathbf{B}_\perp \rangle = 0$.

(d) Verify the Pythagorean identity.

(e) Determine

$$\theta_{Bn} = \arccos\left(\frac{|\langle \hat{\mathbf{n}}, \mathbf{B}\rangle|}{\|\mathbf{B}\|}\right)$$

and classify the geometry using 45° as the conventional boundary between quasi-parallel and quasi-perpendicular.

(f) Construct

$$P_\parallel = \hat{\mathbf{n}}\hat{\mathbf{n}}^T,$$

verify $P_\parallel^2 = P_\parallel$, and show $P_\parallel \mathbf{B} = \mathbf{B}_\parallel$.

2.3 Cauchy–Schwarz from positivity. Let V be an inner-product space.

(a) Starting from

$$0 \le \left\| \mathbf{u} - \frac{\langle \mathbf{v}, \mathbf{u}\rangle}{\|\mathbf{v}\|^2}\mathbf{v} \right\|^2, \qquad \mathbf{v} \neq 0,$$

derive the Cauchy–Schwarz inequality.

(b) Determine the equality condition.

(c) Use Cauchy–Schwarz to prove the triangle inequality.

2.4 Gram–Schmidt in $\mathbb{R}^3$. Apply Gram–Schmidt to

$$\mathbf{v}_1 = (1,1,0), \qquad \mathbf{v}_2 = (1,0,1), \qquad \mathbf{v}_3 = (0,1,1)$$

in the order shown.

(a) Construct an orthogonal basis.

(b) Normalize it.

(c) Form the matrix Q whose columns are the resulting orthonormal vectors and verify $Q^T Q = I$.

2.5 Orthogonal projection onto a subspace. Let

$$W = \operatorname{span}\{(1,1,0), (1,0,1)\} \subset \mathbb{R}^3.$$

Find the orthogonal projection of

$$\mathbf{y} = (2,-1,3)$$

onto W:

(a) by first constructing an orthonormal basis of W;

(b) by solving the normal equations for the least-squares problem.

Show that both methods give the same result and verify that the residual is orthogonal to W.

2.6 Least squares. Find the least-squares solution of

$$\begin{cases} x + 2y = 8, \\ x + y = 4, \\ -x + 3y = 13, \\ x = 0. \end{cases}$$

Write the system as $A\mathbf{x} = \mathbf{b}$, derive the normal equations

$$A^T A\hat{\mathbf{x}} = A^T\mathbf{b},$$

solve them, and verify that

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}.$$

Interpret the last relation geometrically.

2.7 Inner product on a function space. Let $V = C[0,1]$ with

$$\langle f, g\rangle = \int_0^1 f(t)g(t)\,dt,$$

and

$$W = \operatorname{span}\{1,\ 2t-1,\ t^2\}.$$

Use Gram–Schmidt to construct an orthonormal basis for W.

2.8 Complex inner product. On $\mathbb{C}^3$, use the convention

$$\langle \mathbf{u}, \mathbf{v}\rangle = \mathbf{u}^H\mathbf{v}.$$

For

$$\mathbf{u} = (1, \mathrm{i}, 2)^T, \qquad \mathbf{v} = (\mathrm{i}, 1, 1-\mathrm{i})^T,$$

compute

$$\langle \mathbf{u}, \mathbf{v}\rangle, \quad \langle \mathbf{v}, \mathbf{u}\rangle, \quad \|\mathbf{u}\|, \quad \|\mathbf{v}\|.$$

Verify conjugate symmetry and Cauchy–Schwarz numerically.

2.9 Bessel's inequality. Let $\{\mathbf{e}_1, \dots, \mathbf{e}_m\}$ be an orthonormal set in an inner-product space. Define

$$P_m\mathbf{v} = \sum_{k=1}^{m} \langle \mathbf{e}_k, \mathbf{v}\rangle \mathbf{e}_k.$$

Show that

$$\mathbf{v} - P_m\mathbf{v} \perp P_m\mathbf{v}$$

and derive

$$\sum_{k=1}^{m} |\langle \mathbf{e}_k, \mathbf{v}\rangle|^2 \le \|\mathbf{v}\|^2.$$

When does equality hold in the finite-dimensional case?

2.10 Parallelogram law and polarization.

(a) Prove that every norm induced by an inner product satisfies

$$\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2.$$

(b) Show that the ℓ^1 norm on $\mathbb{R}^2$,

$$\|(x,y)\|_1 = |x| + |y|,$$

does not satisfy this identity.

(c) For a real inner-product space, derive the polarization identity

$$\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\left(\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2\right).$$

(d) Explain the significance of the Jordan–von Neumann theorem.

3. Eigenvalues, Diagonalization, Spectral Theory, and Matrix Functions

3.1 Eigenvalues of a linear transformation. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by

$$T\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2a \\ 2b+3c \\ -b-c \end{pmatrix}.$$

(a) Find the matrix of T in the standard basis.

(b) Determine its characteristic polynomial and eigenvalues.

(c) Find a basis for each eigenspace.

(d) Determine whether T is diagonalizable over $\mathbb{R}$.

3.2 Diagonalization and powers of a matrix. For

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix},$$

find P and D such that

$$A = PDP^{-1}.$$

Use the diagonalization to obtain a closed expression for A^n.

3.3 Cayley–Hamilton and the inverse. Let

$$p_A(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$$

be the characteristic polynomial of an invertible matrix A.

(a) Use the Cayley–Hamilton theorem to express A^{-1} as a polynomial in A.

(b) Apply the method explicitly to

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix}.$$

(c) Compare this method conceptually with Gauss–Jordan inversion.

3.4 Hermitian operator and the spectral theorem. Let

$$H = \begin{pmatrix} 2 & 1-\mathrm{i} \\ 1+\mathrm{i} & 3 \end{pmatrix}.$$

(a) Verify that $H^H = H$.

(b) Show that its eigenvalues are real.

(c) Find normalized eigenvectors.

(d) Verify that eigenvectors belonging to distinct eigenvalues are orthogonal.

(e) Construct a unitary matrix U such that

$$U^H H U = D.$$

3.5 Matrix functions through the spectral decomposition. Let A be diagonalizable:

$$A = PDP^{-1}, \qquad D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

(a) Show that for an analytic function f,

$$f(A) = P\ \mathrm{diag}(f(\lambda_1), \ldots, f(\lambda_n))P^{-1}.$$

(b) For

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

compute e^{tA} and interpret the result geometrically.

(c) Verify directly that

$$\frac{d}{dt}e^{tA} = Ae^{tA}.$$

3.6 Quadratic forms. Consider

$$q(x, y, z) = 3x^2 + 2y^2 + 4z^2 + 4xy - 2xz + 6yz.$$

(a) Find a symmetric matrix A such that

$$q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}.$$

(b) Explain why the off-diagonal coefficients of A are half the coefficients of the mixed terms.

(c) State how orthogonal diagonalization of A simplifies the quadratic form.

3.7 Compact self-adjoint viewpoint in L^2. Consider the integral operator

$$(Tf)(x) = \int_0^1 K(x,t) f(t)\, dt$$

with a real symmetric kernel

$$K(x,t) = K(t,x).$$

Assuming all required regularity conditions:

(a) show formally that T is self-adjoint with respect to the L^2 inner product;

(b) explain why the finite-dimensional spectral theorem suggests an expansion in orthogonal eigenfunctions;

(c) identify the analogy between a symmetric matrix and a symmetric integral kernel.

4. Vector Calculus and Tensor Algebra

4.1 Cross product identity. For nonzero vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$ with angle θ, show that, whenever $\mathbf{u} \cdot \mathbf{v} \neq 0$,

$$|\tan\theta| = \frac{\|\mathbf{u} \times \mathbf{v}\|}{|\mathbf{u} \cdot \mathbf{v}|}.$$

Discuss the limiting case $\mathbf{u} \cdot \mathbf{v} = 0$.

4.2 Gradient, divergence, curl, and Laplacian. Let

$$\Phi(x,y,z) = x^2 y + y z^3,$$

and

$$\mathbf{F}(x,y,z) = (xy,\ y^2 - z,\ xz^2).$$

Compute

$$\nabla\Phi, \qquad \nabla^2\Phi, \qquad \nabla \cdot \mathbf{F}, \qquad \nabla \times \mathbf{F}.$$

Then verify explicitly that

$$\nabla \times (\nabla\Phi) = \mathbf{0}.$$

4.3 Vector identities. Using index notation and the Levi–Civita symbol, prove

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

and

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.$$

State the regularity assumptions required for the interchange of partial derivatives.

4.4 Helmholtz-type decomposition. Suppose a sufficiently smooth vector field $\mathbf{F}$ can be written as

$$\mathbf{F} = -\nabla\Phi + \nabla \times \mathbf{A}.$$

Show that

$$\nabla \cdot \mathbf{F} = -\nabla^2\Phi$$

and

$$\nabla \times \mathbf{F} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.$$

What simplification occurs under the gauge condition $\nabla \cdot \mathbf{A} = 0$?

4.5 Elementary tensor algebra: antisymmetry. Let a_{ij} be constants and suppose

$$a_{ij} x^i x^j = 0$$

for every $\mathbf{x}$. Show that

$$a_{ij} = -a_{ji}.$$

You may differentiate with respect to x^k and use

$$\frac{\partial x^\ell}{\partial x^k} = \delta^\ell{}_k.$$

4.6 Tensor products. Let $A^i{}_{kl}$ and $B^j{}_m$ be tensors. Using their transformation laws, show that

$$C^{ij}{}_{klm} = A^i{}_{kl} B^j{}_m$$

is a tensor of rank five. Clearly identify the contravariant and covariant indices.

4.7 Tensor contraction. Let

$$A^{ijk}{}_{lm}$$

be a tensor of rank five. Show from the transformation law that contracting one upper and one lower index,

$$C^{jk}{}_m = A^{ijk}{}_{im},$$

produces a tensor of rank three.

4.8 Metric invariance. Under a coordinate transformation $x^i = x^i(x'^1, \ldots, x'^n)$, show that

$$ds^2 = g_{ij}\, dx^i dx^j$$

is invariant provided

$$g'_{kl} = \frac{\partial x^i}{\partial x'^k} \frac{\partial x^j}{\partial x'^l} g_{ij}.$$

Use

$$dx^i = \frac{\partial x^i}{\partial x'^k} dx'^k.$$

4.9 Prolate spheroidal coordinates. Consider

$$x = a \sinh\eta \, \sin\theta \cos\phi, \qquad y = a \sinh\eta \, \sin\theta \sin\phi,$$

$$z = a \cosh\eta \, \cos\theta,$$

with

$$0 \le \eta < \infty, \qquad 0 \le \theta \le \pi, \qquad 0 \le \phi \le 2\pi.$$

(a) Compute the coordinate basis vectors $\partial\mathbf{r}/\partial\eta$, $\partial\mathbf{r}/\partial\theta$, and $\partial\mathbf{r}/\partial\phi$.

(b) Determine the metric tensor g_{ij}.

(c) Show that the coordinates are orthogonal.

(d) Find the scale factors.

(e) Determine ds^2.

(f) Determine the volume element.

(g) Obtain expressions for the gradient and Laplacian of a scalar field.

4.10 Kinetic energy in curvilinear coordinates. A particle follows $x^i = x^i(t)$ in a curvilinear coordinate system with

$$ds^2 = g_{ij}\, dx^i dx^j.$$

(a) Show that

$$T = \frac{m}{2} g_{ij} \dot{x}^i \dot{x}^j.$$

(b) Compute $\partial T / \partial \dot{x}^i$.

(c) Evaluate

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{x}^i}\right) - \frac{\partial T}{\partial x^i}.$$

(d) Rearrange the result in terms of Christoffel symbols and show how the geodesic acceleration appears.

(e) For a conservative potential $V(x^i)$, with $L = T - V$, derive the Euler–Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}^i}\right) - \frac{\partial L}{\partial x^i} = 0.$$

5. Functions of a Complex Variable

5.1 Complex differentiability and Cauchy–Riemann equations. For each function below, determine where it is complex differentiable and where it is analytic:

$$f_1(z) = z^2, \qquad f_2(z) = \overline{z}, \qquad f_3(z) = |z|^2, \qquad f_4(z) = e^x(\cos y + \mathrm{i} \sin y).$$

Use the Cauchy–Riemann equations and, whenever appropriate, compute $f'(z)$.

5.2 Harmonic functions and harmonic conjugates. Let

$$u(x, y) = x^3 - 3xy^2 + 2x.$$

(a) Verify that u is harmonic.

(b) Find a harmonic conjugate $v(x, y)$.

(c) Construct an analytic function $f(z) = u + \mathrm{i}v$.

(d) Express the result directly as a function of z.

5.3 Complex line integration. Evaluate

$$\int_C z^2\, dz$$

from $z = 0$ to $z = 1 + \mathrm{i}$ along:

(a) the straight line segment;

(b) the broken path $0 \to 1 \to 1 + \mathrm{i}$.

Explain why the results agree.

Then repeat the calculation for

$$\int_C \overline{z}\, dz$$

and discuss why path independence fails.

5.4 Cauchy's integral theorem. Let

$$f(z) = \frac{e^z}{z - 2}.$$

Evaluate

$$\oint_C f(z)\, dz$$

when:

(a) C is the circle $|z| = 1$;

(b) C is the circle $|z| = 3$.

State explicitly which theorem you use in each case.

5.5 Cauchy's integral formula and derivatives. For positively oriented $C : |z| = 2$, evaluate

$$\oint_C \frac{e^z}{z - \mathrm{i}}\, dz, \qquad \oint_C \frac{\sin z}{(z-1)^3}\, dz.$$

Generalize your calculation by stating Cauchy's formula for the nth derivative.

5.6 Taylor series and radius of convergence. Find the Taylor series about $z = 0$ and its radius of convergence for

$$f(z) = \frac{1}{1 - z}, \qquad g(z) = \frac{1}{(1 - z)(2 + z)}.$$

For g, use partial fractions and identify the singularity that determines the radius of convergence.

5.7 Laurent expansions in different annuli. Find Laurent expansions about $z = 0$ for

$$f(z) = \frac{1}{z(z - 1)}$$

in each region:

$$0 < |z| < 1, \qquad |z| > 1.$$

Identify the principal part in each case.

5.8 Classification of isolated singularities. Classify the singularity at $z = 0$ for each function:

$$\frac{\sin z}{z}, \qquad \frac{1 - \cos z}{z^2}, \qquad \frac{1}{z^4}, \qquad e^{1/z}.$$

For removable singularities, determine the value that gives the analytic extension.

5.9 Residues. Compute the residues at all finite isolated singularities of

$$f(z) = \frac{z^2 + 1}{z(z-1)(z+2)}$$

and

$$g(z) = \frac{e^z}{(z - \mathrm{i})^2}.$$

Use at least two different residue techniques across the two examples.

5.10 Residue theorem. Evaluate

$$\oint_{|z|=3} \frac{z^2 + 1}{z(z-1)(z+2)}\, dz$$

for positive orientation. Explicitly identify all poles enclosed by the contour and show how the residue theorem produces the result.

5.11 Real integral by residues. Use contour integration to evaluate

$$\int_{-\infty}^{\infty} \frac{dx}{x^2 + a^2}, \qquad a > 0.$$

Your solution should specify:

(a) the contour;

(b) the enclosed pole;

(c) the residue;

(d) why the contribution from the large semicircle vanishes.

5.12 A pole on the contour. Consider

$$\int_{-\infty}^{\infty} \frac{e^{\mathrm{i}x}}{x}\, dx.$$

The integrand has a pole on the real axis.

(a) Explain why the ordinary residue-theorem prescription cannot be applied without modifying the contour.

(b) Introduce a small semicircular indentation around the origin.

(c) Determine the contribution of the indentation and formulate the result in terms of a Cauchy principal value.

5.13 Branch points and multivalued functions. Consider

$$f(z) = \sqrt{z} \qquad \text{and} \qquad g(z) = \log z.$$

(a) Explain why $z = 0$ is a branch point.

(b) Describe what happens after one complete circuit around the origin for each function.

(c) Choose a branch cut and define a principal branch.

(d) Explain geometrically how a Riemann surface removes the apparent multivaluedness.

5.14 Integration with a branch cut. Let $0 < \alpha < 1$. Consider

$$I(\alpha) = \int_0^\infty \frac{x^{\alpha-1}}{1+x}\, dx.$$

Using a keyhole contour and a consistent branch of $z^{\alpha-1}$:

(a) identify the branch point and branch cut;

(b) relate the integrals above and below the cut;

(c) compute the residue at the enclosed pole;

(d) derive $I(\alpha)$.

6. Integrated Review Problems

The problems in this section intentionally combine several topics. They are especially useful after completing the individual sections above.

R1. Projection, spectral decomposition, and geometry. Let

$$\hat{\mathbf{n}} = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}, \qquad P = \hat{\mathbf{n}}\hat{\mathbf{n}}^T.$$

(a) Prove $P^2 = P$ and $P^T = P$.

(b) Determine all eigenvalues and eigenspaces of P.

(c) Diagonalize P orthogonally.

(d) Show that

$$e^{tP} = I + (e^t - 1)P.$$

(e) Interpret $\mathrm{Im}(P)$ and $\ker(P)$ geometrically.

R2. A matrix-valued rotation problem. Let

$$J = \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}.$$

(a) Show $J^2 = -I$.

(b) Compute $e^{\theta J}$ from its power series.

(c) Show that $e^{\theta J}$ is orthogonal and has determinant 1.

(d) Interpret it as an active transformation.

(e) Determine the corresponding passive coordinate transformation.

R3. Differential, gradient, and metric. In polar coordinates (r, θ),

$$ds^2 = dr^2 + r^2 d\theta^2.$$

Let

$$\Phi(r, \theta) = r^2 \cos(2\theta).$$

(a) Write the covector $d\Phi$.

(b) Use the inverse metric to obtain the contravariant components of $\nabla\Phi$.

(c) Express $\nabla\Phi$ in the orthonormal polar basis $\{\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta\}$.

(d) Compute $\nabla^2\Phi$ and interpret the result.

R4. Complex analysis and two-dimensional potential theory. Let

$$f(z) = z + \frac{1}{z}, \qquad z \neq 0,$$

with

$$f(z) = \Phi(x, y) + \mathrm{i}\Psi(x, y).$$

(a) Determine Φ and Ψ.

(b) Verify the Cauchy–Riemann equations away from $z = 0$.

(c) Show that Φ and Ψ are harmonic away from the singularity.

(d) Find the singularity and residue of f.

(e) Explain the relationship between analytic functions and pairs of harmonic scalar fields in two dimensions.

R5. Tensor and coordinate invariance. Let $\mathbf{v}$ be a vector and ω a covector.

(a) Write their transformation laws under a general change of coordinates.

(b) Prove that $\omega_i v^i$ is invariant.

(c) Given a metric g_{ij}, define $v_i = g_{ij} v^j$ and prove that $v_i v^i$ is invariant.

(d) Relate the result to the invariant line element $ds^2 = g_{ij} dx^i dx^j$.

Final Study Checklist

Before Exam 1, you should be able to:

- move fluently between matrices, systems of equations, and linear maps;
- use determinant, rank, kernel, image, and rank–nullity correctly;
- perform and interpret changes of basis, including dual-basis transformations;
- work with real and complex inner products, orthogonality, Gram–Schmidt, and orthogonal projections;

- connect least squares to orthogonal projection;
- determine eigenvalues/eigenvectors, test diagonalizability, and use the spectral theorem in the Hermitian/symmetric case;
- compute and interpret elementary matrix functions;
- use gradient, divergence, curl, and Laplacian and prove standard vector identities;
- manipulate tensor indices, transformation laws, contractions, and metrics;
- test analyticity using the Cauchy–Riemann equations;
- evaluate complex contour integrals using Cauchy's theorem/formula;
- construct Taylor and Laurent expansions and classify singularities;
- compute residues and apply the residue theorem;
- handle the conceptual issues associated with poles on contours, branch points, branch cuts, and Riemann surfaces.

Solutions are intentionally not included. Complete derivations and clear mathematical reasoning are part of the preparation.